

Vignesh Sivakumar

Principal Software Engineer

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PROFESSIONAL SUMMARY

Principal Software Engineer with 12+ years of expertise in distributed systems, search grid architectures, and cloud platform optimization. A technical leader with a track record of driving large-scale system modernization, scaling low-latency services, and building high-performing engineering teams. Passionate about domain-driven system design, ML feature platforms, and operational excellence.

TECHNICAL SKILLS

Programming Languages: Java, Scala, Golang, Python, Bash

AI & Agentic Systems Engineering: Autonomous AI Agents, Multi-Agent Frameworks (LangGraph, CrewAI, Google ADK, Strands, OpenAI Agents SDK), Model Context Protocol (MCP), RAG Architectures, Generative AI Feature Generation

Technologies & Frameworks: Spring Boot, RESTful APIs, gRPC, Scalatra

Databases: PostgreSQL, MySQL, DynamoDB

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitLab CI/CD, Jenkins

System Design & Architecture: Microservices, Event-Driven Architecture, Message Brokers (Kafka, SQS), Distributed Caching (Redis, Memcached), Indexing & Search (Lucene, OpenSearch), Distributed Storage (Hadoop)

Testing & Debugging: JUnit, Postman, Swagger, k6

Security & Authentication: OAuth, JWT, SAML, OpenID Connect

Version Control & Collaboration: GitLab, GitHub, Agile, Scrum, Kanban

EXPERIENCE

PRINCIPAL SOFTWARE ENGINEER | 09/2025 to 07/2026

Socure India Private Limited - Chennai (Remote)

- Spearheaded the end-to-end architectural vision and system design to decompose heavy legacy monolith workflows into modular, lightweight, and high-performance microservices, establishing scalable APIs, domain data models, and RiskOS-native integration pathways.
- Architected and directed the technical deprecation roadmap for legacy ID+ systems, proactively surfacing observability blind spots and dependency risks across teams to ensure a zero-downtime transition to next-generation platforms.

- Mentored senior and staff engineers to advance career growth, standardized design and code review workflows across organization, and provided architectural guidance to unblock cross-functional teams.
- Led efforts to enhance system resilience by strengthening automated CI/CD deployment pipelines and acted as technical lead during high-severity global infrastructure incidents, including AWS STS outages and Shai-Hulud security vulnerabilities.
- Engineered enterprise-grade shared software libraries, including ultra-low-latency distributed caching module and robust protocol buffer validation frameworks, to improve service stability and developer velocity.

STAFF SOFTWARE ENGINEER | 09/2023 to 09/2025

Socure India Private Limited - Chennai (Remote)

- Designed the multi-parameter fuzzy search OpenSearch query engine for Anti-Money Laundering (AML) screening, scaling it to handle 2,500+ QPS with sub-150ms latency while passing 98% of name-degradation match tests.
- Led the structural re-architecture of the core monolithic service into decoupled modular components, significantly cutting production incidents and enabling system capacity to scale from 350 QPS to 3,000 QPS.
- Architected OpenSearch cluster performance tuning, shard allocations, and indexing strategies, driving a 67% reduction in total infrastructure costs without compromising query SLAs.
- Overhauled real-time ML feature computation mechanism, reducing redundant feature calculations by 90% per request and increasing overall platform throughput while cutting compute runtime costs.
- Rearchitected internal rate-limiting system into language- and platform-agnostic, no-code solution, eliminating developer overhead for enabling rate limits on APIs and workflows.
- Architected low-code external vendor integration framework, reducing onboarding and integration engineering effort by 50%.
- Utilized Generative AI tools to design and launch a new product within 48 hours, auto-generating over 200 production ML features with zero operational bugs.
- Designed an elastic search layer leveraging Amazon DynamoDB for cost optimization and architected ephemeral testing environments for isolated, automated pre-production deployments.

SENIOR SOFTWARE ENGINEER II | 09/2022 to 09/2023

Socure India Private Limited - Chennai (Remote)

- Rearchitected internal asynchronous request-auditing pipelines, eliminating database locks and queuing bottlenecks, resulting in over 50% increase in system processing throughput.
- Developed a priority-aware batch job orchestration system enabling data science teams to submit heavy analysis workloads while enforcing request-rate limits.

SENIOR SOFTWARE ENGINEER I | 01/2021 to 09/2022

Socure India Private Limited - Chennai (Remote)

- Co-designed a low-code framework for machine learning feature generation, reducing feature coding time by over 65% and accelerating time-to-market.
- Redesigned ML model hot-reloading mechanisms to smooth out memory usage during updates, reducing latency spike occurrences by over 97%.
- Optimized core ETL framework data pipelines and batch processing mechanisms, accelerating throughput from 67,000 requests/hour to over 1,000,000 requests/hour (~15x increase).
- Stabilized legacy auditing databases by implementing critical updates and query optimizations, preventing storage exhaustion every 4 hours and ensuring long-term production stability.

- Developed vendor API call tracking and telemetry service, generating automated usage reports for Finance to enhance budget planning and improve client billing accuracy.

MEMBER OF TECHNICAL STAFF | 06/2014 to 12/2020

Zoho Corporation Private Limited - Chennai

- Designed and implemented a high-concurrency, multi-tenant distributed search grid capable of ingesting and querying millions of documents daily across multiple enterprise applications.
- Engineered automated primary data center detection and multi-backup region replication mechanisms, ensuring instant failover and continuous data availability during regional outages.
- Authored and merged patches into Apache Hadoop codebase, fixing critical edge-case bugs and enhancing overall system stability.
- Engineered custom data locality logic for Hadoop to optimize processing relative to server states, building queue consumers to efficiently process and index millions of documents daily.
- Designed an offline web search framework using browser IndexedDB for network-disconnected usage, and built metrics collection frameworks for real-time cluster health and query monitoring.

EDUCATION

Anna University - Chennai, India | Bachelor of Technology

Information Technology, 05/2014

GPA: 9.56/10